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The effect of the particle size on Fischer–Tropsch synthesis for ZSM-5 zeolite supported cobalt-based catalysts†

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A series of mesoporous ZSM-5 zeolite supported cobalt-based catalysts with the cobalt crystal sizes in the range of 4.5–18.1 nm were prepared for syngas conversion. The highly selective synthesis of various liquid fuels including gasoline, jet fuel and diesel range hydrocarbons is achieved with different cobalt nanoparticle sizes.

Fischer–Tropsch synthesis (FTS) is an important catalytic transformation route for converting syngas (a mixture of H₂ and CO, obtainable from coal, natural gas, biomass, *etc.*) into C₂₊ hydrocarbons, where the products are environmentally friendly with a very low content of sulfur compounds.¹ Generally, the product selectivity of conventional FTS follows the Anderson–Schulz–Flory (ASF) distribution, which is not selective toward a specific range of products (such as C₅–C₁₁ gasoline, C₈–C₁₆ jet fuel and C₁₀–C₂₀ diesel).² Improving the selectivity of a specific fuel by breaking the ASF distribution is more cost- and energy-efficient and has become a hot research topic recently, but it remains a great challenge.³

The cobalt-based catalysts have the advantages of high activity and high selectivity for long-chain hydrocarbons.⁴ Recently, bifunctional catalysts have been developed to couple cobalt with a secondary hydrocracking component to form one catalyst for direct conversion of syngas to a specific range of liquid fuels.⁵ Most research studies focused on zeolite-supported catalysts, especially using the ZSM-5 zeolite as the support.⁶ It was found that the modification of the pore

architecture of ZSM-5 can enhance the catalytic performance and control the hydrocarbon distribution by improving the mass transfer and decreasing the diffusion limitation during the reaction.⁷ In addition to the pore structure, the acid sites, especially the Brønsted acid sites, on ZSM-5 can also tune the hydrocarbon distribution through secondary hydrocracking and isomerization of hydrocarbons formed on cobalt particles, and further improve the gasoline selectivity.⁸

It is well known that the properties of metallic cobalt, especially the cobalt particle size, can also affect the FTS activity and product distribution.⁹ De Jong *et al.* reported that a larger cobalt particle size (6–16 nm) for Co/CNFs was beneficial for the FTS activity and C₅₊ selectivity while a smaller cobalt particle size (<6 nm) tended to be blocked by a large amount of irreversibly adsorbed CO leading to a lower activity.¹⁰ However, Borg *et al.* found that the C₅₊ selectivity exhibited a volcano-like curve with respect to the cobalt particle size in the range of 3–18 nm for Co/Al₂O₃ catalysts.¹¹ The influence of the cobalt particle size was also investigated for some other catalyst systems (Co/SiO₂, Co/TiO₂, Co/ZrO₂, *etc.*).¹² Nevertheless, to the best of our knowledge, the effect of the cobalt particle size on the product distribution on FTS over the bifunctional Co/ZSM-5 catalysts has not been reported. Herein, hierarchical ZSM-5 (Z5) supported Co-*x*/Z5 catalysts, where *x* represents the diameter of the reduced cobalt particle size ranging from 4.5 to 18.1 nm, were fabricated to unveil the cobalt crystallite size effect on the product distribution over the bifunctional Co-*x*/Z5 catalysts.

Z5 was prepared by treating the pristine H-ZSM-5 using TPAOH solution (ESI†). Fig. 1 displays the texture, morphology and microstructure of the Z5 zeolite. The XRD pattern (Fig. 1a) suggests that the crystalline structure of the Z5 zeolite (PDF #44-0003) is not destroyed after TPAOH treatment and the N₂ adsorption–desorption curve of the Z5 zeolite with a hysteresis loop (Fig. 1b) reveals its mesoporous structure. The SEM image (Fig. 1c) shows that the crystalline structure of ZSM-5 was partially broken by the mesopores formed after TPAOH

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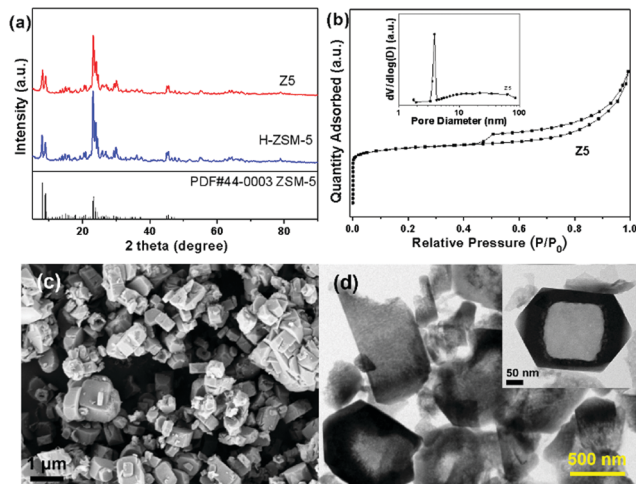


Fig. 1 (a) XRD patterns, (b) N_2 adsorption-desorption curve, and (c) SEM and (d) TEM images of the Z5 zeolite.

treatment, and the TEM image (Fig. 1d) further shows that there exists a large void inside the Z5 zeolite, which favors the mass transport.¹³

Using the Z5 zeolite as the support, we tuned the cobalt particle size by varying the cobalt loading as well as the drying and calcination processes (ESI[†]). The mean size of the cobalt particles of the reduced Co-*x*/Z5 catalysts increases from 4.5 to 18.1 nm, as determined from TEM images (Fig. S1 and Table S1, ESI[†]). The HRTEM images reveal that an interplanar distance of 0.205 nm is ascribed to the Co(111) plane (Fig. S1, ESI[†]). In addition, the fresh Co-*x*/Z5 catalysts are composed of Co_3O_4 (PDF #43-1003) and Z5 from the XRD patterns in Fig. S2a (ESI[†]). The average Co_3O_4 and Co particle sizes were also calculated by the Scherrer equation and the following equation: $d(Co) = 0.75d(Co_3O_4)$, respectively (Table S1, ESI[†]). The cobalt particle sizes obtained from XRD are consistent with the TEM analysis. Meanwhile, the relative crystallinity of Z5 decreased with the increase of the Co particle size, indicating that the framework structure was partially destroyed after cobalt loading. The N_2 adsorption-desorption experiment shows that with an increasing Co particle size, the specific surface area and pore volume of Co-*x*/Z5 catalysts decrease from 312 m² g⁻¹ and 0.36 cm³ g⁻¹ for Co-4.5/Z5 to 273 m² g⁻¹ and 0.30 cm³ g⁻¹ for Co-9.9/Z5, respectively (Fig. S2b and Table S1, ESI[†]). Conversely, when the Co particle size further increases to 18.1 nm, the specific surface area increases to 297 m² g⁻¹ with little change in the pore volume (Fig. S2c, ESI[†]), suggesting that the larger cobalt particles hardly enter the pore channel of Z5 and tend to locate on the surface of the Z5 zeolite. The mesopore volume of Co-*x*/Z5 catalysts decreases gradually from 0.29 cm³ g⁻¹ for Co-4.5/Z5 to 0.21 cm³ g⁻¹ for Co-18/Z5, while the micropore volume changes slightly correspondingly (Table S1, ESI[†]), indicating that mesopores are blocked partly by a large number of cobalt species for the sample with the larger Co particles.

The acidic properties of Co-*x*/Z5 catalysts were measured through pyridine-adsorbed Fourier-transform infrared spectroscopy (Py-FTIR) and NH_3 -temperature programmed desorption (NH_3 -TPD).

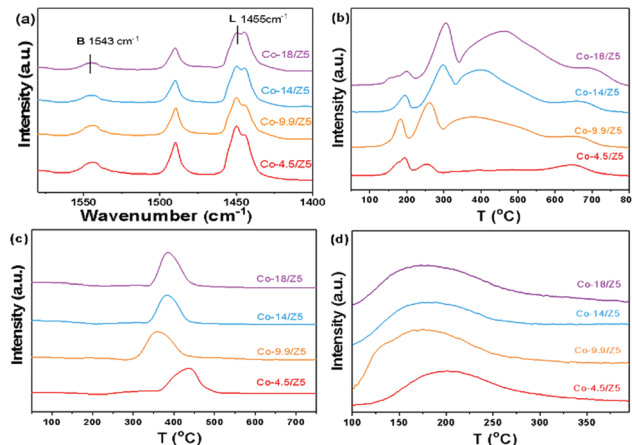


Fig. 2 (a) Py-FTIR, (b) H_2 -TPR, (c) CO-TPD and (d) H_2 -TPD of Co-*x*/Z5 catalysts.

The Py-FTIR peaks at 1455 and 1543 cm⁻¹ can be ascribed to the Lewis and Brønsted acid sites, respectively (Fig. 2a). It is found that the intensity of both the acid sites decreased gradually from 443.7 and 141.5 μmol g⁻¹ for Co-4.5/Z5 to 316.9 and 80.1 μmol g⁻¹ for Co-18/Z5, respectively (Table S2, ESI[†]), though these Co-*x*/Z5 catalysts were all prepared with the same Z5 support. Therefore, the acidity of the Co-*x*/Z5 catalysts depends on the cobalt nanoparticle size, which may be related to the interaction between cobalt species and the surface groups of Z5. NH_3 -TPD (Fig. S2d, ESI[†]) showed two NH_3 desorption peaks at 250 and 400 °C, which can be attributed to the hydrogen-bonded NH_3 and the NH_3 adsorbed on the Brønsted acid sites, respectively.¹⁴ The intensity of both the acid sites decreases with increasing cobalt particle size, which correlates well with the Py-FTIR results. The largest amount of acid sites is obtained for the Co-4.5/Z5 sample due to its high crystallinity as shown in Fig. S2a (ESI[†]).¹⁵

The H_2 -temperature programmed reduction (H_2 -TPR) technique was performed to study the reduction of different Co particle sizes, and the profiles of the Co-*x*/Z5 catalysts are presented in Fig. 2b. All the catalysts exhibited four peaks, which corresponded to the decomposition of the residual nitrate (150–200 °C),¹⁶ the reduction of Co_3O_4 to CoO (200–350 °C) and further reduction to metallic Co (350–600 °C) and the reduction of cobalt silicate or cobalt aluminate species, which were formed by the surface cobalt species interacting with the Si-OH or Al-OH groups of Z5 (600–700 °C).¹⁷ It is observed that the second peak shifts to a higher temperature with increasing cobalt crystallite size. The third peak appears obviously for Co-9.9/Z5 and also shifts to a higher temperature with a further increase in the cobalt crystal size to 18.1 nm. The results reveal that the interaction between cobalt and the Z5 zeolite is enhanced with larger cobalt particles, which may further reduce the crystallinity of Z5 as shown by the XRD patterns (Fig. S2a, ESI[†]) and decrease the acidity of the Co-*x*/Z5 catalyst.

The X-ray photoelectron spectroscopic (XPS) studies showed that all the Co-*x*/Z5 catalysts exhibited four peaks (Fig. S3, ESI[†]). Two peaks at binding energies (BEs) of 780.3–780.9 eV and 796.2–796.7 eV are attributed to Co 2p_{3/2} and Co 2p_{1/2} peaks, and another two peaks at BEs of around 786.4 and 803.1 eV are

ascribed to the corresponding satellite peaks, respectively.¹⁸ The BEs of Co 2p_{3/2} and 2p_{1/2} increased from 780.3 and 796.2 eV for Co-4.5/Z5 to 780.9 and 796.7 eV for Co-18/Z5, which suggests the enhanced interaction between Co and Z5 with the increasing Co nanoparticle size, consistent with the H₂-TPR results.¹⁹

To further understand the adsorption behaviors of the CO and H₂ reactants on Co-*x*/Z5 catalysts, the CO and H₂-TPD experiments were performed (Fig. 2c and d). All the Co-*x*/Z5 catalysts show one CO and one H₂ desorption peak in the range of 300–500 °C and 100–300 °C, respectively, which are assigned to the chemisorbed CO and H₂, respectively.²⁰ It is obvious that when the cobalt nanoparticle size increases from 4.5 to 9.9 nm, the CO and H₂ desorption peaks shift to a lower temperature, but they move reversely to a higher temperature as the particle size further increases to 18.1 nm. The lower CO and H₂ desorption temperatures are beneficial for their activation, leading to a higher FTS activity. Meanwhile, we also note that all the catalysts exhibit similar CO adsorption capacity, while the amount of H₂ adsorption first increases significantly from 22.6 μmol g⁻¹ for Co-4.5/Z5 to 124.9 μmol g⁻¹ for Co-9.9/Z5, and then somewhat decreases to 77.9 μmol g⁻¹ for Co-18/Z5 (Table S2, ESI†). The degree of cobalt reduction and its dispersion were also calculated by H₂-TPD and O₂ titration (Table S1, ESI†). The degree of cobalt reduction increases from 55.2 to 92.0% as the cobalt crystal size increases from 4.5 to 9.9 nm and then remains the same around 90% for the larger cobalt nanoparticles. With the increasing Co crystallite size, its dispersion decreases gradually from 19.8 to 5.2%. Additionally, the average cobalt particle size calculated by Co dispersion is in line with the XRD and TEM results.

The FTS activity and the product distribution of all the Co-*x*/Z5 catalysts are presented in Fig. 3, and Fig. S3 and Table S3 (ESI†). The CO conversion increases gradually from 16.8 to 58.7% with the increase in the cobalt particle size from 4.5 to 9.9 nm (Fig. S3, ESI†), which is mainly due to the enhanced cobalt reduction with the appropriate cobalt dispersion. The lower degree of cobalt reduction for smaller cobalt particles, especially for Co-4.5/Z5, inhibits its complete reduction to metallic cobalt in syngas conversion. However, with the Co nanoparticle size further increasing to 18.1 nm, the CO conversion decreases to 50.0%, which may be due to the very low cobalt dispersion. Besides, Co-9.9/Z5 exhibits the lowest CO desorption temperature among all studied catalysts, which favors the dissociation and the conversion of CO.²¹ In addition to the CO conversion, the product distribution is also influenced by the cobalt particle size. Typically, with increasing Co crystal size, the selectivities towards CH₄ and C₂–C₄ gradually decrease and the Co-4.5/Z5 catalyst shows the highest CH₄ and C₂–C₄ selectivities of 28.3% and 15.2%, respectively (Fig. 3a). For liquid range hydrocarbons, the selectivities towards gasoline and jet fuel range hydrocarbons first increase to reach maximum values of 44.5% and 43.5% with the cobalt crystallite sizes of 9.9 and 14.4 nm, respectively, and then gradually decline to 40.1 and 41.8% with the further increase in the cobalt size to 18.1 nm (Fig. 3b and c). Moreover, the selectivity towards the diesel range hydrocarbons increases linearly with the cobalt particle size and reaches 35.4% for Co-18/Z5 (Fig. 3d and Fig. S4, ESI†).

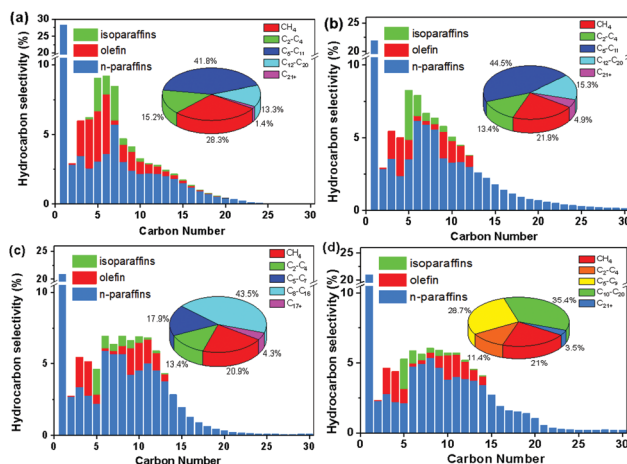
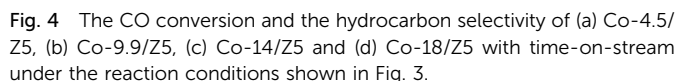


Fig. 3 The detailed hydrocarbon distribution of (a) Co-4.5/Z5, (b) Co-9.9/Z5, (c) Co-14/Z5 and (d) Co-18/Z5. Reaction conditions: 1.0 g of catalyst, *T* = 240 °C, *P* = 2.0 MPa, H₂/CO = 2, WHSV = 3000 mL g_{cat}⁻¹ h⁻¹ and time-on-stream of 48 h.

As shown in Fig. S5 and Table S3 (ESI†), the chain growth probability (α value) of Co-4.5/Z5 with a much smaller Co nanoparticle size is 0.67, significantly lower than those of 0.77–0.81 for the other samples, leading to higher selectivity towards undesired CH₄ and C₂–C₄ hydrocarbons in FTS. Similar results over the traditional Co-based FTS catalysts were also observed by other researchers.²² In addition, Co-4.5/Z5 possesses the highest amounts of Brønsted acid sites as determined by Py-FTIR, which may result in the over-cracking of the long-chain hydrocarbons to C₁–C₄ hydrocarbons. For the Co-18/Z5 catalyst, the larger cobalt particles tend to locate on the surface of Z5, and the micropore volume and the number of Brønsted acid sites are the lowest among all the catalysts, which reduces the chance for the produced long-chain hydrocarbons to interact with the Brønsted acid sites in the channel of Z5, thus decreasing the secondary hydrocracking reaction to produce heavier hydrocarbons, especially for the diesel range fuel and wax. High gasoline and jet fuel range hydrocarbon selectivities are obtained for Co-9.9/Z5 and Co-14/Z5, because they possess an appropriate cobalt crystal size and an appropriate amount of Brønsted acid sites, which benefit the C–C bond formation on metallic cobalt and the hydrocracking of primary hydrocarbons on the Z5 zeolite.

The stability of the Co-*x*/Z5 catalysts is also studied, and the CO conversion and hydrocarbon distribution with time-on-stream on the four catalysts are presented in Fig. 4. The C₅⁺ selectivity for the Co-*x*/Z5 catalysts remained stable with a slight decrease in CO conversion during a 72 h stability test. For Co-4.5/Z5 and Co-9.9/Z5 (Fig. 4a and b), the selectivity towards gasoline range hydrocarbons decreased slightly from 42.5% and 44.4% at 24 h to 41.9% and 43.9% at 72 h, respectively. The C₈–C₁₆ jet fuel selectivity remained at 43.3% for Co-14/Z5 after 72 h of the stability test (Fig. 4c), indicating the good stability for jet fuel production from syngas. For the Co-18/Z5 catalyst, the C₁₀–C₂₀ selectivity remained above 35% after 72 h of syngas conversion (Fig. 4d).



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There are no conflicts to declare.

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